Fatma Coruh Ergün Bilgisayar Müh.
CSE 225

CSE225 DATA STRUCTURES **Student Name:**_____
Borahan Tümer **MIDTERM EXAM** **Student Number:**_____

1. This exam includes 5 questions.
2. This is an **open-book and open-notes exam.**
3. Time for this exam is **2 hours.**
4. Do not forget to **write down your name and number on each page**.
5. Please answer the questions in empty spaces left below the questions! You may use the back of your papers if necessary.
6. You may get partial points. **Work neatly and show all your work!**
7. Good luck!!! ☺

Kitabe-i Seng-i Mezar

I
Hiç bir şeyden çekmedi dünyada
Nasırdan çektiği kadar;
Hatta çirkin yaratıldığından bile
O kadar müteessir değildi;
Kundurası vurmadığı zamanlarda
Anmazdı ama Allah'ın adını,
Günahkar da sayılmazdı
Yazık oldu *Süleyman Efendi*'ye

II
Mesele falan değildi öyle,
To be or not to be kendisi için;
Bir akşam uyudu;
Uyanmayıverdi.
Aldılar, götürdüler.
Yıkandı, namazı kılındı, gömüldü.
Duyarlarsa öldüğünü alacaklılar
Haklarını helal ederler elbet.
Alacağına gelince...
Alacağı yoktu zaten rahmetlinin.

III
Tüfeğini deppoya koydular,
Esvabını başkasına verdiler.
Artık ne torbasında ekmek kırıntısı,
Ne matrasında dudaklarının izi;
Öyle bir ruzigar ki,
Kendi gitti,
İsmi bile kalmadı yadigar.
Yalnız şu beyit kaldı el yazısıyla:
"Ölüm Allah'ın emri,
Ayrılık olmasaydı."
_____**Orhan Veli Kanık**

Cımbızlı Şiir
Ne atom bombası,
Ne Londra Konferansı;
Bir elinde cımbız,
Bir elinde ayna;
Umurunda mı dünya!
_____**Orhan Veli Kanık**

Vatan İçin...
Neler yapmadık şu vatan için!
Kimimiz öldük;
Kimimiz nutuk söyledik.
_____**Orhan Veli Kanık**

Hay Kay...
Yosun kokusu
Ve bir tabak karides
Sandıkburnu'nda
___**Orhan Veli Kanık**

Gemliğe Doğru...
Gemliğe doğru
Denizi göreceksin;
Sakın şaşırma!
_____**Orhan Veli Kanık**

Anadolu.....
Beşikler vermişim Nuh'a,
Salıncaklar, hamaklar;
Havva Anan dünkü çocuk sayılır,
Anadoluyum ben.
Tanıyor musun?
_____**Ahmed Arif**

.....
Geçme namert köprüsünden,
Ko aparsın su seni.
Yatma tilki gölgesinde,
Ko yesin aslan seni.
Diyarbakırlı Sait Paşa

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Q1. (25pts)
Consider the following piece of code!

```
...
long res=0;
...
for (i=1; i<=n; i++)
  for (j=1; j<=i; j*=2)
    res += (i+j);
printf("res=%d", res);
...
```

Figure 1A: Code for part (a)

```
...
long res=0;
...
for (i=1; i<=n; i++)
  for (j=1; j<=i; j*=2)
    res += (i+j)*(j mod i < 4);
printf("res=%d", res);
...
```

Figure 1B: Code for part (b)

a) (10pts)
What is the value of res, **in terms of *n***, printed out at the printf statement in Fig. 1A? **Show all your work!**

$$S = \sum_{i=1}^{n} \sum_{j=1}^{\lfloor \log_2 i \rfloor + 1} i + 2^{j-1}$$

b) (15pts)
What is the value of res, **in terms of *n***, printed out at the printf statement in Fig. 1B? **Show all your work!**

Q2. (20pts)
Consider the following structures of data:

```
struct LL_node {
    double info;                /* 4 bytes */
    struct LL_node * next;      /* 4 bytes */
}

struct BST_node {
    double info;                /* 4 bytes */
    struct BST_node * left;     /* 4 bytes */
    struct BST_node * right;    /* 4 bytes */
}
```

You are to decide on whether to keep n data nodes declared as above in a linked list (LL) or in a binary search tree (BST). The performance measure you are to use to compare the performance of a LL and a BST is defined as follows:

Performance = Total memory space allocated * Average Execution Time

or

$$P = M * T.$$

a. (10pts)
In terms of n, what should be **the maximum possible execution time of a BST** T_{max} so it can outperform a LL (i.e., $P_{BST} < P_{LL}$) ?

b. (10pts)
Assume that the data are presented to your program in such an order that the first 7 nodes (or less) compose a full (or a complete) binary tree and all subsequent nodes (i.e., all nodes starting from 8 up until n) are located on the same path in the BST. Find **the interval for the integer values of n** such that, for the BST explained above, $P_{BST} < P_{LL}$?

Q3. (20pts)

Consider the arithmetic string below:

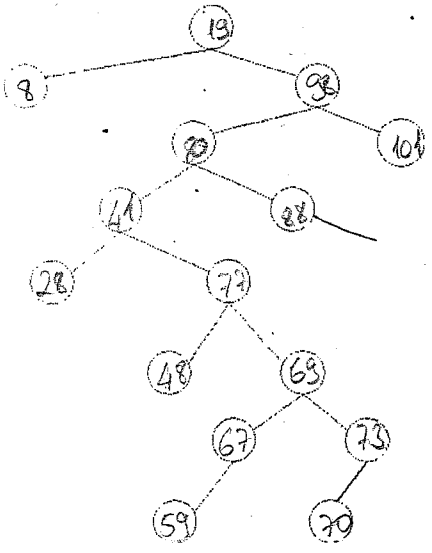
3 1 + 2 3 2 ^ a 2 / 4 3 - b - c * 2 4 1 + ^ 3 4 * 2 2 * - / 6 2 * - + = 0

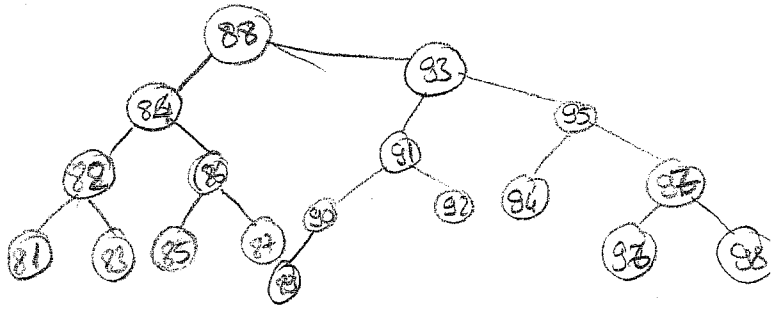
where a, b and c are arithmetic operators or positive integers. Assuming the results of all subsequent operations turn out integers, find a, b and c!

Show all your work!

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Q4. (40pts)
S = {8, 19, 28, 41, 48, 59, 67, 69, 70, 73, 77, 80, 88, 98, 101}





Consider the set of numbers S as given above (i.e., above is only shown what elements there exist in S. The sequence of numbers above is not important). Initially, this set consists of 15 integers.

a) (5pts)
Fill all numbers in the above binary tree topology such that it becomes a binary search tree (BST).

b) (9pts) (3+3+3)
What is the **worst case, best case, and average case search times of the above BST**, assuming that a comparison of the searched item with the key of a tree node is 1 time unit?

c) (12pts) (6+6)
Assuming no key duplication, what is the **highest number of nodes** that can be added to a subtree whose root is the right child of the node with key 88 without relocating (i.e., changing the places of) existing keys? Please **provide the set of numbers that you can add**!

d) (14pts)
In how many different ways can you insert all nodes in part (c), provided that

- $t_{av} \leq 5.75$, and
- $t_{wc}^{(d)} \leq t_{wc}^{(b)} + 1$

$t_{wc}^{(b)}$ is the worst case search time in part (b) of the question

$t_{wc}^{(d)}$ is the worst case search time in part (d) of the question

Q5. (20pts)
Consider the following piece of code:

```
int sum(int n){
    if (n==0)
        return 0;
    else return n^2 + sum(n-1);
}
```

Figure 5A.

```
int sum(int n){
  if (n==0)
    return 0;
  else return pow(0.5,n)+sum(n-1);
}
```

Figure 5B.

a) (10pts)
In terms of *n*, what does *sum* Fig. 5A return to its original caller function?

b) (10pts)
In terms of *n*, what does *sum* Fig. 5B return to its original caller function?